

Lesson 10 Proofs in Number Theory

You may use any theorems in Lesson 10.

1. Let a, b, c be positive integers. Prove or disprove: If a and b are relatively prime, and b and c are relatively prime, then a and c are relatively prime.
2. Are there integers x and y such that $7x + 9y = 1$? If there are, find them. If not, explain why not.
3. Let a, b be integers. Prove or disprove: If $2|ab$ and a is odd, then b is even.
4. Disprove: Let a, b, c be integers, if $a|c$ and $b|c$, then $ab|c$.
5. Prove that if a and b are relatively prime, and $a|c$ and $b|c$, then $ab|c$.
6. Let a and b be integers. Use Theorem 2 of Lesson 10 to prove that if ab and $a + b$ are relatively prime, then a and b are relatively prime.